

Discrete Mathematics (MATH 208)

Fall 2022

Friday, 11/14/2022

Homework 11: Chapters 28, 29

Score: / 10

Name (Print): _____

Chapter 28 - The Two Fundamental Counting Principles

1 pts

1. My piggy bank contains 20 pennies, 4 nickels, 7 dimes, and 2 quarters. In how many ways can I select one coin?

2 pts

2. My piggy bank contains 20 pennies, 4 nickels, 7 dimes, and 2 quarters. In how many ways can I select four coins, one of each value?

- 1 pts 3. Code words consist of five letters followed by five digits. How many code word contain at least one X ?

- 2 pts 4. Code words consist of five letters followed by five digits. How many code word contain exactly one X ?

Chapter 29 - Permutations and Combinations

- 1 pts 1. In how many ways can the ten digits $0, 1, 2, \dots, 8, 9$ be written in a row?

1 pts

2. How many bit strings of length ten have exactly four 0's?

2 pts

3. How many bit strings of length ten have at most four 0's?

2 (bonus)

4. In how many ways can a subset of two numbers from 1 to 100 (inclusive) be selected if the selected numbers cannot be consecutive?